

Darknet traffic detection and characterization with models based on decision trees and neural networks [☆]

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ARTICLE INFO

Keywords:

Traffic classification
Darknet
Security
Deep web
Neural networks
Benchmark

ABSTRACT

The Darknet is a set of networks and technologies, having as fundamental principles anonymity and security. In many cases, they are associated with illicit activities, opening space for malware traffic and attacks to legitimate services. To prevent Darknet misuse is necessary to classify and characterize its existing traffic. In this paper, we characterize and classify the real Darknet traffic available from the CIC-Darknet2020 dataset. In that sense, we performed the feature extraction and grouped the possible subnets with an n-gram approach. Furthermore, we evaluated the relevance of the best features selected by the Recursive Feature Elimination method for the problem. Our results indicate that simple models, like Decision Trees and Random Forests, reach an accuracy above 99% on traffic classification. Our methodology represents a gain of up to 13% in comparison with the state-of-the-art.

1. Introduction

Internet presents a diversified niche segregated by the level of availability and anonymity of the services. One of the Internet layers is the *Surface Web*, this reduced part is the one widely available by search indexes and commonly used by users looking for so-called common services and applications. The *Deep Web*, a set generally “encrypted” and sought after by those looking for peculiar services, makes available the results not indexed by conventional search engines. This set has its principles based on the continuous change of hosting and establishing connections through secure peers, Peer-to-Peer (P2P). the *Darknet* is a subset of the *Deep Web* that expands these principles and further restricts peer connection (Mirea et al., 2019).

Darknet demonstrates the highest level of security techniques to ensure the anonymity of groups and service providers, preserving the identity of the subjects involved in the relationships. Examples include selling products on the black market, negotiating services, and exchanging information. Although these are some of the activities that are on the side of illegality, *Darknet* demonstrates a heterogeneous network established on principles founded on privacy in the midst of sharing

digital content. In a general way, the objective of the *Darknet* is to carry out secure communication between peers, preserving the confidentiality and completeness of their interactions and keeping the anonymous sharing nature. With this, the Darknet becomes a secure repository for any individual to establish activities regardless of their nature, ensuring the advantage of difficult traceability.

As these properties from the *Darknet* are highly used for malicious purposes, the malicious traffic from the *Darknet* is potentially dangerous for computer networks. So, the classification of the traffic origin and categorization of the type of application in these cryptographic situations is one way to prevent damage from *Darknet* traffic, and it is one of the objectives of the study of the *Traffic Classification* problem. The objective of determining certain classes analyzing the data traffic, i.e., checking connection duration patterns, information about the origin and destination of this data, connected ports, and the type of application related to the analyzed flow is useful for many security applications. For instance, in intrusion detection, *Quality of Service* management in scalability, preventing the spread of *malware* or (Parchekani et al.) service attacks. The methods available in the literature vary and

[☆] This work has been partially supported by the Department for drug policies of the Presidency of the Council of Ministers (Italian Government) through the project InstradaME (CUP F49I20000100001) and the National Operational Programme Metropolitan Cities 2014-2020(PON METRO) through the project MeSm@rt (CUP F41I18000230006).

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may be linked to the analysis of ports and *payload* or inserting statistical inferences to classify the record under analysis (Medeiros et al., 2019).

It is possible to find few efforts to detect and characterize *Darknet* traffic using Deep Learning (DL), such as *Deep Image Learning* (Gurdip Kaur & Habibi Lashkari, 2020), there are notable efforts to identify traffic from the same set using traditional machine learning techniques. This demonstrates that state-of-the-art is being developed for complex systems, but there is still the need for studies evaluating the efficiency of simpler models.

In this regard, the present work approaches the traffic classification problem of *Darknet* using a representative database – the *CIC-Darknet2020* (Gurdip Kaur & Habibi Lashkari, 2020). The main contributions and results of our work can be summarized as follows:

- Higher-performing machine learning models to the *traffic classification* area in *Darknet*, with an improvement of up to 13% concerning the state-of-the-art. Furthermore, we have shown that complex models are not always necessary for the addressed tasks;
- Is shown the importance of careful analysis and preprocessing of the attributes to identify how data quality can be improved and, consequently, the performance of the models created with them;
- From the analysis of the existing fields, new features are created aiming for the best performance of the traffic origin detection and the characterization of the application;
- We analyze the results of the models created with the Decision Tree (DT), Random Forest (RF), and Multilayer Perceptron (MLP) to verify their applicability in the proposed problems and to compare them through the *McNemar* statistical test to determine the existence of a statistically significant difference between them;
- Finally, we also perform a feature selection process with the algorithm *Recursive Feature Elimination* to analyze the features' relevance in their respective tasks. The validation resulted in an accuracy of 99.89% for the traffic source classification task using only 30 attributes. For the application categorization task, with 50 attributes, the resulting accuracy was 98.62%, indicating that removing a significant part of these characteristics does not affect the overall performance.

The remainder of this article is structured as follows: works related to traffic classification using different approaches are discussed in Section 2. Section 3 describes the *dataset* used for classification and characterization of traffic on *Darknet*, and describes the pre-processing steps that we adopted to improve the quality of the data used in the training of models. In addition, we also describe the extracted attributes based on those existing in the *dataset*. The creation and performance analysis of the DT, RF, and MLP, models are described in Section 4. In addition, we also perform a model selection by estimating their performance and comparing them with statistical tests in Section 5, and we perform a feature selection to analyze the importance of attributes in traffic classification tasks. Finally, the conclusions and future work of this article are presented in Section 6.

2. Related work

Before showing different solutions to the *Traffic Classification* problem, it is necessary to understand the definition of the problem, which consists of using traffic data between sender and receiver to classify and categorize the application used. One of the main challenges is to carry out this task using encrypted data, an approach made in two databases provided by the *University of New Brunswick*, the *ISCXVPN2016* (Draper-Gil et al., 2016) and *ISCXTor2016* (Lashkari et al., 2017), which, respectively, provide traffic over networks using Virtual Private Network (VPN) and The Onion Router (Tor).¹ Other related works, like (Bovenzi

et al., 2020) and (Montieri et al., 2020), can be cited, focusing on the correlation between privacy-preserving and traffic classification.

Recently, in work published by Gurdip Kaur and Habibi Lashkari (2020), a database was made available that is the union of the other two mentioned above, called *CIC-Darknet2020*. This work performs the classification of applications from *Surface Web* and *Darknet*, respectively, which is defined as the benign origin and *Darknet*. In this case, the *Darknet* traffic was encrypted using VPN and Tor. In addition to publishing the database, the author presented an accuracy of 92% for identifying the origin of the traffic and 86% for its categorization in their classification model using deep neural networks, using a technique called *Deep Image Learning*. Something important to note is that although the database provides information about the traffic source and destination Internet Protocol (IP)s, it is not possible to detect its real origin. It occurs since the traffic that passes through Tor makes its way through intermediary networks in order to hide the user's real location. As for the traffic coming from VPNs, the detection of the origin is not possible because they are fake IPs or *bogons*.

In Draper-Gil et al. (2016), the authors address the classification of the traffic via VPN on the dataset *ISCXVPN2016* in two stages with the usage of neural networks. The first, using MLP as the activation function for the second stage, a *Recurrent Neural Network* to identify the six classes employed by the *dataset*. Establishing two application categorization scenarios, the authors use the K-Nearest Neighbors (KNN) and C4.5 DT models for the task. They present an accuracy above 80%, emphasizing the C4.5, which had better precision measurement results.

In Lotfollahi et al. (2020), using the same dataset, the author presents the *Deep Packet framework* approach for the traffic classification problem. The *framework* comprises two deep learning methods, a convolutional neural network, and an autoencoder, both for the classification and characterization tasks. With accuracy and precision results above 90%, the work played an important role in supporting other methods besides the one used to solve the problem. In Crotti et al. (2007) authors propose alternative approaches for the classification problem by using ports, not recommended as it has low classification rates, *payload* inspection, and the use of a statistical approach. It demonstrates an accuracy of around 91% for the classification of Hypertext Transfer Protocol (HTTP), Post Office Protocol (POP3), and Simple Mail Transfer Protocol (SMTP) protocols and 87% for File Transfer Protocol (FTP), Internet Message Access Protocol (IMAP), Secure Shell (SSH), and TELNET Protocol (TELNET).

The works of (Lotfollahi et al., 2020) and (Draper-Gil et al., 2016), although they have contributed to the categorization of the application by traffic, are not focused on data from *Darknet*. Gurdip Kaur and Habibi Lashkari (2020) addresses categorization with *deep learning* in two layers, the first related to origin classification and the second in relation to traffic coming from *Darknet*, checking the attributes of the *dataset* most important for ranking. Even with an accuracy of 86% for the problem, the authors do not compare it with simpler classification models.

Different from the aforementioned works, our proposal uses simple and easy-to-apply models, such as DT, RF, and MLP, in order to compare the results of the traffic origin classification models with the categorization of the application of data from *Darknet*. Furthermore, we verify the influence of the creation of new attributes by comparing the models to verify whether there is a significant difference in performance. Through this comparison, the models used, although simple, allowed results close to 100% of overall accuracy, unlike the *Deep Learning* models found in state-of-the-art. Thus, the database provided by Gurdip Kaur and Habibi Lashkari (2020) is chosen to achieve the proposed objectives, expanding it with new features, inserting information on the source and destination IP addresses, dividing them into *n-grams* (Wress-

¹ <https://www.torproject.org/>.

Table 1
Related work summary.

Proposal	Dataset	Attributes	ML/DL	Acc.
Gurdip Kaur and Habibi Lashkari (2020)	CIC-Darknet2020	VPN, Tor	Deep Image Learning	86%
Draper-Gil et al. (2016)	ISCXVPN2016	VPN	MLP, KNN, C4.5	80%
Lotfollahi et al. (2020)	ISCXVPN2016	All	CNN + AutoEncoder	90%
Crotti et al. (2007)	CIC-Darknet2020	HTTP, POP3, SMTP, FTP, IMAP, SSH, TELNET	Statistical	91%
This work	CIC-Darknet2020	All + <i>n</i> -grams	DT, RF, MLP	99.89%

negger et al., 2013), and manipulating the original records.² These changes and comparisons allow the evolution of both the base and the highlighted theme (Table 1).

3. Dataset and methodology

3.1. Dataset

The CIC-Darknet2020 dataset is utilized for the development of classification models. It has records of traffic from the surface web and from both the datasets ISCXTor2016 and ISCXVPN2016 (Gurdip Kaur & Habibi Lashkari, 2020), containing Tor and VPN traffic data, respectively. CIC-Darknet2020 considers as Darknet traffic the data that comes from Tor and VPN networks. The dataset establishes two classification tasks related to the traffic classification problem. The first task is the origin detection of the traffic, in which the classes are Benign, representing surface web traffic, and Darknet. Fig. 1 shows the data distribution regarding the traffic origin. The other task is the characterization of the traffic in 8 labels. These labels are the applications that originate the traffic, being them: browsing, email, chat, audio-streaming, video-streaming, file-transfer, VoIP (Voice Over IP), and P2P (Peer-to-Peer).

The dataset has a variety of fields for analysis, bringing information about the origin and destination IPs, ports, duration of packet traffic, and other related measures. There are 141.528 records, of which 24.310 come from the Darknet and 117.218 were from benign networks. This proportion difference comes from the fact that the services found on the Darknet are not indexed by search engines. Besides, they need special applications, like Tor, to access them (Gurdip Kaur & Habibi Lashkari, 2020). Fig. 2 shows the distribution of the applications that originate the traffic. It is clear that streaming, audio, and chat applications are the most common for data from Darknet, while for benign networks, they are the minority.

3.2. Data preparation

The performance of data-based models is greatly affected by the quality of the data used during their creation. This challenge increases when using large *datasets* from the real world, which usually has quality-related problems due to missing, inconsistent data. There are several reasons for the low quality of *datasets* in the real world, which can be caused by human or computer error, incorrect data submitted by users, and faulty data collection instruments, among other problems that may occur. Data pre-processing techniques aim to improve data quality and may improve the efficiency of subsequent processes (Han et al., 2011). The following subsections present the pre-processing techniques applied in this paper to improve the learning efficiency and the accuracy of machine learning models in the tasks of detection and characterization of *Darknet* traffic.

² CIC-Darknet2020 Internet Traffic dataset (Gurdip Kaur & Habibi Lashkari, 2020) records bi-directional flows represented by a flow-id and its attributes, e.g., IP Address, Src Port.

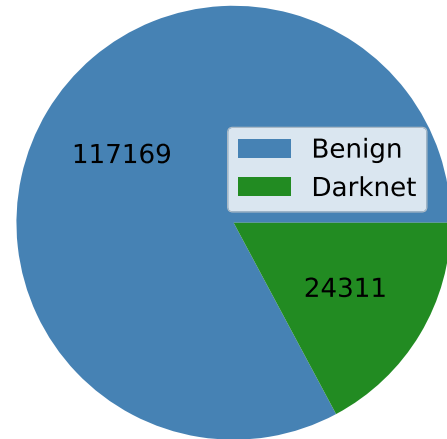


Fig. 1. Distribution by the data source.

3.2.1. Labels correction

The dataset comes with duplicate labels on both label types made available by the dataset. Through label inspection, it is possible to observe a lack of standardization on label names and redundancy. To approach this issue, we standardize the names by choosing one of the duplicates. For example, records with the labels AUDIO-STREAMING (capital letters) were replaced by Audio-Streaming.

3.2.2. Features encoding

Most machine learning models only work on numeric data. However, frequently datasets include categorical variables representing, as the name says, categories or labels. Unlike numerical variables, values on a categorical variable can not be ordered; that is, its magnitude is not relevant to the task in question. The categories usually are not numeric; therefore, it's necessary to apply an encoding method to transform them into numbers. One of the possibilities is the simple mapping from each category to integers; however, the resulting numbers can turn out to be orderable between them, something that is not desirable for category values (Zheng & Casari, 2018).

The value of a subnet regarding another is irrelevant to the problem, so features corresponding to IP addresses on the dataset are examples of categorical variables. Since the value of a subnet does not necessarily indicate the origin of the traffic or its generating application. The concept of *n*-grams models approaches the problem of IP address encoding with a more generic mapping. Initially, these models were proposed for natural language processing and currently are the dominant representation in many detection systems (Wressnegger et al., 2013).

One of the possible applications of *n*-gram models is the direct capture of IP address subnets. The RFC 950 (Mogul et al., 1985) defines a 3-level interpretation model for internet addresses, where the highest level represents the internet as a whole, the level below are the individual networks, and the last level represents the subnets that are useful for networks belonging to moderately large organizations. Thus, each of these IP addresses interpretation levels can be represented with the utilization of unigrams, bigrams, and trigrams models, in an attempt to capture the traffic subnet of origin.

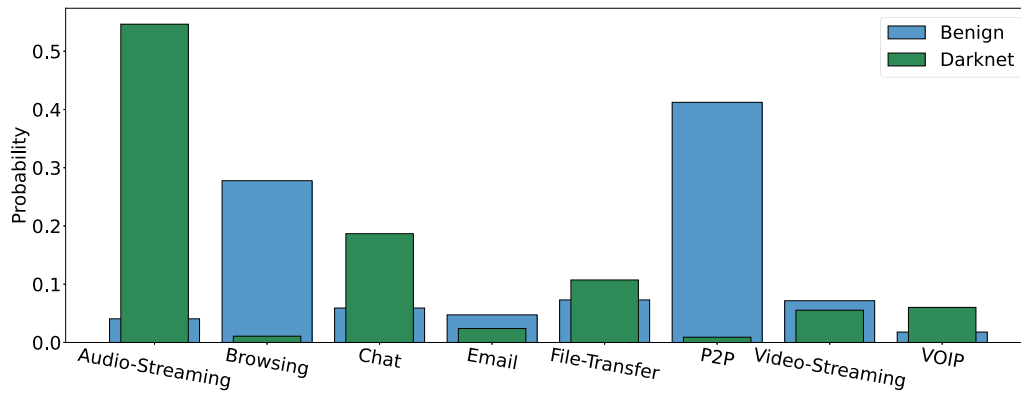


Fig. 2. Occurrence probability of the traffic related to service.

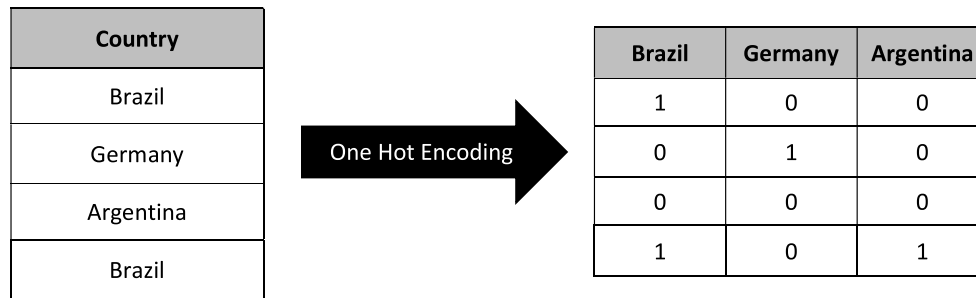


Fig. 3. Feature encoding with One Hot Encoding.

Table 2

IPs addresses division in grams example.

IP	Unigram	Bigram	Trigram
172.168.15.14	172	172 168	172 168 15
182.170.224.79	182	170 224	182 170 224

Table 2 exemplify the division of the IP addresses in grams, allowing us to observe how the process is similar to what the subnet mask does for subnet identification but, as we do not have access to the subnet masks, we use the grams for each possible interpretation level. With these new attributes, a machine learning model can be able to learn the most common prefixes or subnets for a traffic type as a piece of relevant information for new traffic predictions.

The first encoding is applied to the unigrams, bigrams, and trigrams created from the IP addresses. We used the hashing encoding technique to create 100 new features named with the prefix *col_* followed by the attribute ID generated by the method. As noted by Weinberger et al. (2009), the hashing encoding technique enables attributes encoding regarding one hot encoding. One hot encoding does categorical feature encoding by transforming each of the categories into new binary features indicating if that category belongs to the corresponding dataset example, as shown in Fig. 3. This could lead to a large number of new features depending on the number of unique categories on a categorical variable. On the other hand, the hashing encoding has the disadvantage that the new features can not be easily interpreted as the generated features can not be mapped back to their original values. Furthermore, as we could extract the country of origin from the network addresses, this information is used as a feature for model training after applying an ordinal encoding transformation.

One of the most important transformations applied to data is feature scaling. Generally, few machine learning models have good performance with features with different scales. Therefore, it is necessary to transform feature values to have the same numeric range. For this

purpose, we applied the standard scaling method to all features. This technique works by subtracting the feature mean from values and scaling them, so the features have a unitary variance (Zheng & Casari, 2018).

3.2.3. Feature extraction

The dataset provides information about the origin and destination IP addresses, enabling the extraction of more features related to the network traffic. One of the possibilities is the application of one-hot encoding. However, the usage of n-grams can help in decreasing the percentage of false positives on predictions. Hence, the dataset was extended using the IP addresses, fragmenting them into unigrams, bigrams, and trigrams. Furthermore, were used information from hosting, geolocation, and bogons (false addresses), between other features with the aid of the *IpInfo* library.

Another feature extracted was the time that the traffic data capture started from the traffic *timestamp*. Fig. 4 shows the relationship between the capture time for both labels related to the origin of the traffic using *TCPDump* and *Wireshark*. It is possible to verify that two different patterns take place in the traffic generation behavior for each label, being possible to observe that these patterns on traffic time can be used by the models in the learning phase, making the traffic time a relevant feature for traffic classification.

Besides being possible to see the timetable distinction, this time relation allows seeing when there is a high probability of utilization of each network. For the *Benign* network, can be observed a high traffic density between 7 am, and 12 am, with some peaks at dawn. On the *Darknet* traffic side, the traffic distribution is sparser, having a high utilization peak at dawn and two smaller peaks in the afternoon. These relations are almost an exclusive disjunction; that is, there is a considerable probability of not having high traffic on the *Darknet* at the same time of *Benign* networks, which allows smoother learning of these patterns on classification models.

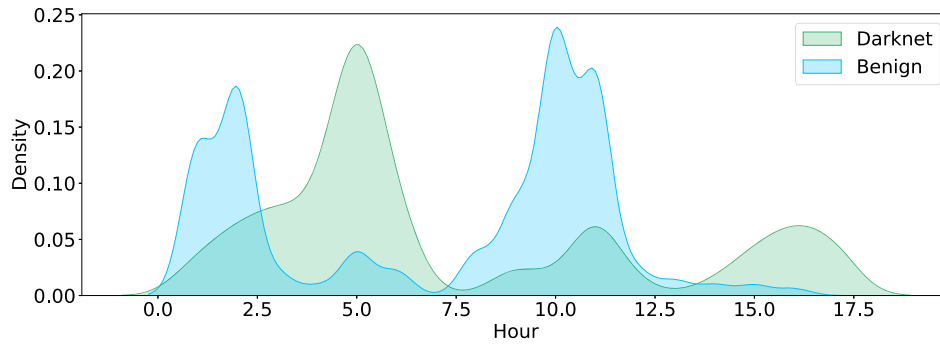


Fig. 4. Traffic density in a given hour.

Fig. 5. K-fold illustration with $k = 5$ (Fedotenkova, 2016).

3.3. Model evaluation

With the variety of classification models available, metrics are needed to assess their performance in a given task so that it is possible to choose the best model. This process is called model selection. Two common methods for accuracy assessment are the holdout and the random subsampling methods. The holdout method works by randomly partitioning two independent sets, training and a test set, and the accuracy is defined as the number of hits on the test set. The random subsampling is similar to the holdout method, but the holdout method is repeated k times, and the accuracy is the average of the accuracies obtained in each iteration (Han et al., 2011).

To estimate the accuracy of the models, stratified k -fold cross-validation is used where the k -fold has $k = 10$. This method divides the data into k groups of approximately equal size and distribution of labels, where the first *fold* is used as the validation set, and the method is trained on the remaining $k - 1$ folds. Unlike the holdout and random subsampling methods, here, each sample is used the same number of times for training and once for testing. Accuracy with k -fold is defined as the average accuracy of all training and validation pairs combinations; Fig. 5 illustrates the procedure. Precision, *recall* and *F-score* (Géron, 2019) are used as metrics in both classification tasks. The equations of the evaluation metrics used are listed below, where TP, TN, FP, and FN are the classification of true positives, true negatives, false positives, and false negatives, respectively.

- Precision: can be thought of as a measure of exactness, i.e., the percentage of examples labeled as positive are actually such;

$$precision = \frac{TP}{TP + FP} \quad (1)$$

- Recall: is a measure of completeness, i.e., the percentage of positive samples labeled as such, the same as sensitivity;

$$recall = \frac{TP}{TP + FN} \quad (2)$$

- F-score: can be viewed as a weighted average of the precision and recall. It reaches its best value at one and the worst at zero. Both precision and recall have the same contribution to the F-score.

$$F - score = \frac{2 \times (precision \times recall)}{precision + recall} \quad (3)$$

One of the ways to approach the model selection problem is by selecting the model with fewer errors using k -fold. However, k -fold only provides an estimate of the error in the actual population of the future data. Although the model errors appear to be different between them, this difference may not be statistically significant. To determine whether the difference in the error means it is statistically significant, we need to employ a statistical significance test. Thus, we can state that the means cannot vary outside a confidence interval with future samples (Han et al., 2011).

The choice of the statistical test must be made carefully, to avoid misinterpretation of the models. In (Dietterich, 1998), five commonly used statistical tests are compared with the objective of determining the one with the slightest error of the type I, that is, the one with the most negligible probability of detecting a difference when it does not exist. The authors conclude that for algorithms that can be executed only once, the use of the *McNemar* test is recommended. However, for algorithms that can be executed ten times, they recommend the 5x2 cv test (*cross-validation*) as it has the smallest error of type I.

4. Evaluation

The evaluated models are based on decision trees and a simple neural network selected due to their simplicity and ease of interpretation. In addition, it conducts a selection of characteristics to estimate the importance of attributes according to their influence on classification. Next, the chosen models (Géron, 2019) are briefly described.

- **Decision Trees (DT):** non-parametric supervised learning models that can be used for classification and regression. It works by learning simple decision rules inferred from the data to predict the target variable. They are simple models to understand and interpret, and the generated trees can be visualized. Some advantages of decision trees are: Simple to understand and interpret, and possible to be visualized. Non-require data normalization or hard work for data preparation. Its cost is logarithmic in the number of data points used to train the tree. Moreover, it is considered easy to interpret and explain by a boolean logic supported by a white-box model. However, the disadvantages of decision trees include the following: DT can create over-complex trees that do not generalize the data well. This is called overfitting. It can be unstable, considering its small variations in the data might result in a completely different tree being generated. By the way, it is possible to mitigate almost all of these problems. Fig. 6 presents a DT with green

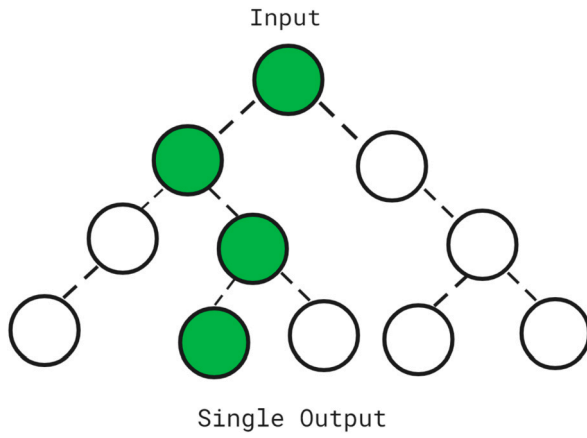


Fig. 6. Typical architecture of a DT model.

nodes selected by the decision tree algorithm based on boolean logic. The entropy concept is used by DT, where the p_+ is the probability of a positive class, in contrast to the p_- probability of a negative class in a subset of the training example, as is possible to see on $E(S) = -p_{(+)} \log p_{(+)} - p_{(-)} \log p_{(-)}$.

- **Random Forest (RF)**: is a *ensemble* model that uses DTs as weak classifiers in order to generate a robust classifier. The RF trains each of the DTs with the *Bagging technique* to generate a classifier with better performance than its individual components. RFs are frequently used as “black box” models in computational intelligence solutions. They generate reasonable predictions across a wide range of data while requiring trim configuration. However, RFs, in general, are more complex compared to DTs. In this case, RF is an excellent way to conduct more extensive experiments with more data, even losing the comfortable model interpretability offered by DT. Fig. 7 shows a set of DTs used to create the RF model.
- **Multilayer Perceptron (MLP)**: is a class of feedforward artificial neural networks (ANN). It is composed of an Input layer, one or more layers of perceptrons, also called Hidden layers, and an Output layer responsible for the final classification. Except for the input layer, every MLP node is a perceptron model and a non-linear activation function. Fig. 8 shows the architecture of an MLP. Its advantages are adaptive learning, a huge number of possible applications, and do not assume the underlying probability density functions or other probabilistic information about the pattern classes under consideration compared to other probability-based models. But, its disadvantages can be mentioned by the hidden layers with a non-convex loss function where more than one local minimum exists. Therefore, different random weight initializations can lead to different validation accuracy and require tuning several hyperparameters, such as hidden neurons, layers, and iterations. To conclude, MLP is sensitive to feature scaling. Keeping it in mind, the MLP can be a good neural network for any problem, mainly for temporal series.

However, a disadvantage of the models used is that they cannot be trained *online*. In other words, they cannot learn from a new example unless the model is retrained with all the previous data and the new examples. All experiments were done with the *sklearn* library, in the Python programming language, on a computer with a *Intel Core i5-7200U* processor with 4 2.5 GHz cores, 20 GB of RAM, and the *Ubuntu 20.04* operating system. Each one of the models were trained keeping the default parameters defined by *sklearn*.

4.1. Darknet traffic detection

In Figs. 9a, 9b, and 9c corresponding to the DT, RF, and MLP models it is possible to observe the models confusion matrices on the task of

Table 3

Models evaluation metrics summary.

	Precision	Recall	F-score	10-fold
DT	0.9964	0.9971	0.9967	99.89%
RF	0.9987	0.9947	0.9967	99.9%
MLP	0.9992	0.9915	0.9953	99.84%

Table 4

Confusion matrix labels.

Initials	Label	Initials	Label
AS	Audio Streaming	P	P2P
B	Browsing	FT	File Transfer
VS	Video Streaming	C	Chat
V	VOIP	E	Email

traffic detection. It is possible to observe that, for both classes, the vast majority of them are correctly classified. Through the confusion matrices of Fig. 9 it is possible to see that for the class *Darknet*, there is a percentage of 0.45% wrongly classified as *Benign*, and for the opposite, 0.02%. In an application in which the correct detection of *Darknet* traffic is more important, MLP is the most suitable as it has a lower number of false positives among the analyzed models.

The Table 3 summarizes the metric values of each model in the classification between the labels *Benign* and *Darknet*, with the values in bold as the best results obtained for comparison between the models. Furthermore, the model achieved an accuracy of 99.89% in *10-fold* and good generalization, thus better results than the literature, where Gurdip Kaur and Habibi Lashkari (2020) achieved an accuracy of 94% in the detection of *Darknet* traffic.

4.2. Darknet traffic characterization

Fig. 11 lists, in polar coordinates, the complement values of the precision, recall, F-score, and also the classification error for the DT models, RF and MLP respectively, for traffic characterization. The complements of these metrics were presented to provide better visualization. It is possible to verify that only the class *Browsing* had a discrepant distance from the maximum value of the metrics. In addition, among all models, DT had the best results in all metrics, while MLP and RF had worse values in more labels, in addition to *Browsing*, RF also had worse results in relation to *Video-Streaming*, while MLP had worse results on *P2P*, *Email*, *Video-Streaming* and *Browsing* tags. The confusion matrices of Figs. 10a, 10b, and 10c, respectively corresponding to DT, RF, and MLP, and the acronyms of the rows and columns correspond to the Table 4 labels, it is evident that the common errors are related to traffic with *Chat* and *Audio-Streaming* labels, which may indicate that there is some similarity in the traffic corresponding to these labels, which can cause confusion in the models.

Fig. 11d shows the errors obtained by the models in the classification of each type of service associated with the traffics, making it evident that the classes that obtained more significant classification errors are those least represented in the *dataset*. RF and MLP attenuated the errors of these classes. Therefore, they are more suitable models for traffic classification, assuming that the addition of new examples follows the same training set probability distribution and that the traffic corresponding to these labels is not of great importance for the application of the model. We obtained a minor classification error with the DT even in classes with less representation. So, the model proposed by Gurdip Kaur and Habibi Lashkari (2020) using *deep learning* obtained an overall accuracy of 86% against 99.03% of our DT's model. Considering it, we achieved an improvement of 13.03%, which may indicate that the problem of characterization of *Darknet* traffic is better solved

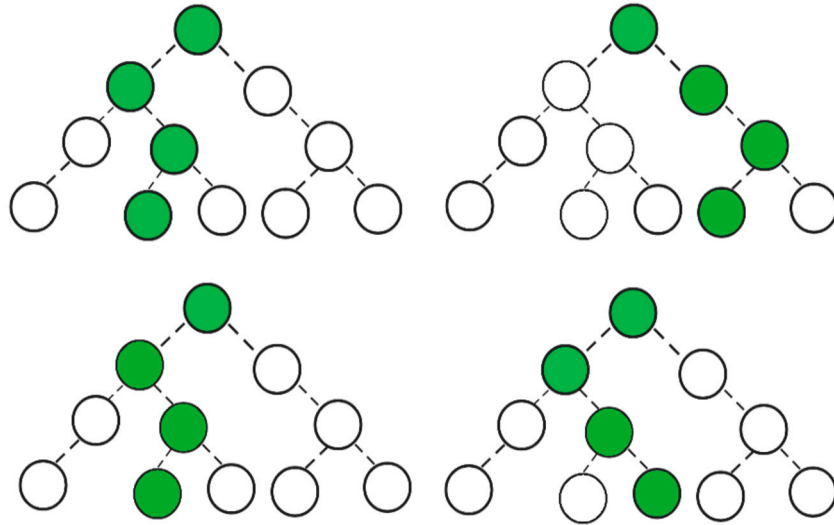


Fig. 7. Typical architecture of an RF model.

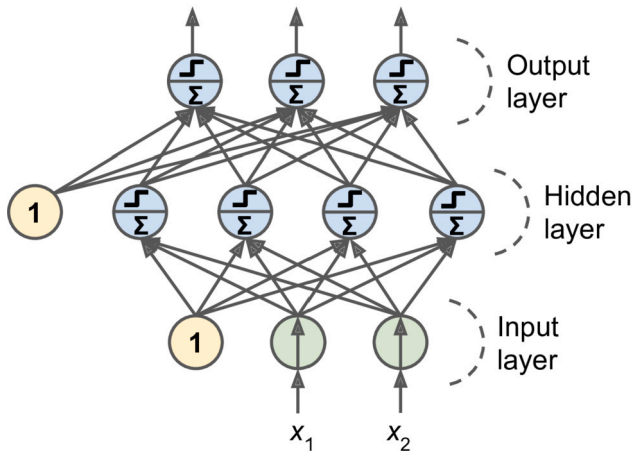


Fig. 8. Typical architecture of an MLP model (Géron, 2019).

with simpler and easier-to-interpret models without the need to resort to more complex models and that demands more computing resources. However, in applications where it is necessary to update the model as new examples arrive, MLP is the most suitable model as it is the only one that allows *online* learning.

5. Evaluation

5.1. Model evaluation

As the models trained in this paper can be run ten times, we chose 5x2 cv as the test for model selection. In 5x2 cv, there are five executions of cross-validation with 2 folds. In each execution, the data is randomly partitioned into two sets of approximately equal size, where each of the models is trained on one set and tested on the other. After the results are generated, a paired *student's t* test is performed to generate the final statistics (Dietterich, 1998).

Table 5 summarizes the accuracies estimated by the 10-10-fold with all models for the traffic detection and characterization tasks. The accuracies are estimated as the average of the accuracies obtained in all 10-fold runs. As it is possible to observe, all models obtained very similar results for traffic detection. For the characterization, the decision tree had better results than the random forest and the multilayer perceptron. However, although there are noticeable differences in the

Table 5

10-10-fold estimated accuracy for darknet traffic detection and characterization.

	DT	RF	MLP
Detection	99.905	99.907	99.837
Characterization	99.154	98.754	97.731

Table 6

p-values for traffic detection and characterization models comparison.

	DT vs RF	DT vs MLP	MLP vs RF
Detection	0.095	0.286	0.389
Characterization	0.013	0	0.001

performance of the models, it is necessary to run the 5x2 cv to see if these results are statistically significant.

Figs. 12a and 12b summarize the accuracy of the models in the detection and characterization tasks of *Darknet* traffic in boxplots. It is evident that in both classification tasks, the DT obtained better results. We also observed that the MLP had a greater variance in the estimates than the other models. To verify whether the difference in the accuracy estimates of the models is statistically significant, we conduct statistical tests.

As a null hypothesis for the statistical tests, we consider that the models do not significantly differ. We also assume that the confidence margin for the statistical test result is 95%. If the p-value is less than the α threshold of 0.05, the value used as a convention (Dahiru, 2008), then the null hypothesis is rejected, and there is a statistically significant difference between the compared models. Table 6 the p-values are summarized by comparing the models from 2 to 2 to see if there is any difference between them. The results indicate that there is no statistically significant difference between the models for detecting *darknet* traffic, while all models for characterization have a statistically significant difference between them.

Despite the results favoring DT as the best model for both classification tasks, essential points must be considered. The DT is a model that does not allow us to update it after its training. Every time a sample or a set of them arrives, it is necessary to retrain the entire model to consider the new samples. As the problem of interest consists of data that can come as a stream, it would be interesting to update the model as new samples appear. This type of update is called online learning. Thus,

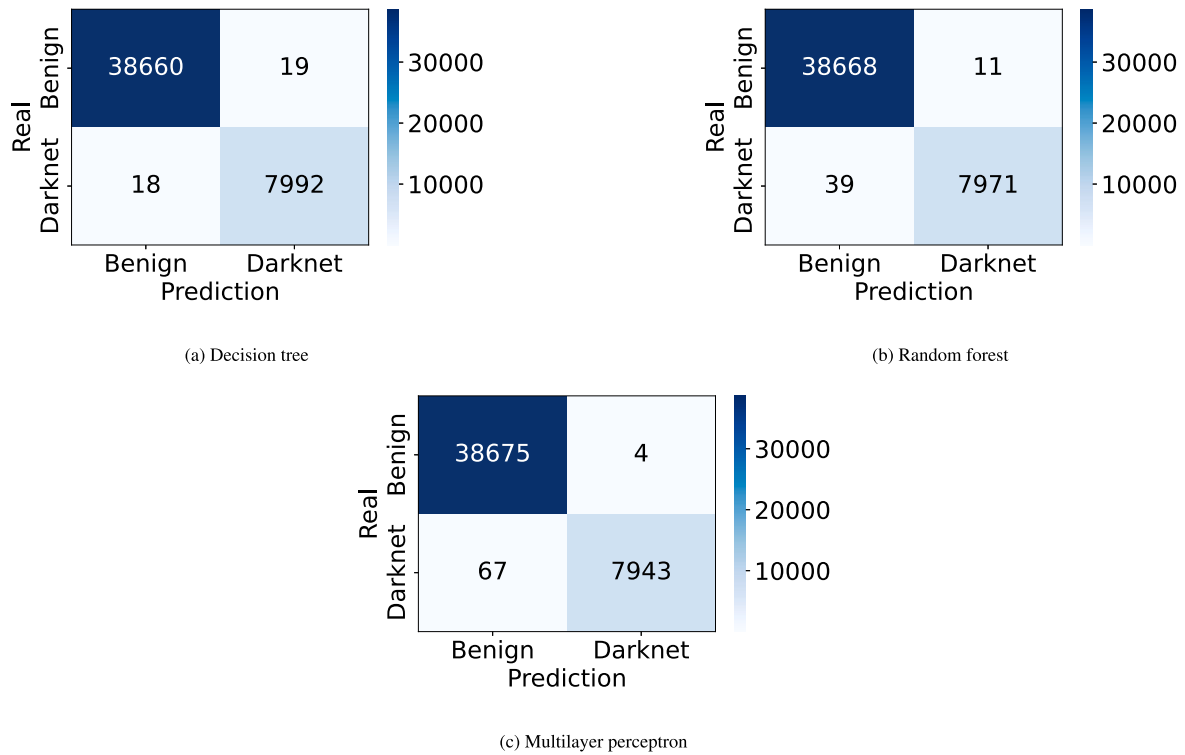
Fig. 9. Confusion matrices of *Darknet* traffic detection.

Table 7
Summary of attribute selection results.

	# attributes	Final validation	10-fold
Detection	28	99.95%	99.91%
Characterization	73	99.12%	98.94%

Table 8
Characterization.

Attribute	Relevance
col_91	0.7628
col_49	0.1205
Bwd Init Win Bytes	0.0418
col_24	0.0408
Idle Min	0.0141
col_96	0.0034
Idle Std	0.0021
col_45	0.0018
hour	0.0017
Average Packet Size	0.0015
Flow IAT Std	0.0011
col_1	0.0009
Idle Mean	0.0009
Flow IAT Min	0.0008
Fwd Packet Length Max	0.0008
FIN Flag Count	0.0007
Src Port	0.0006
FWD Init Win Bytes	0.0006
Fwd IAT Total	0.0005
Flow Duration	0.0004
col_71	0.0004
Fwd Packets/s	0.0004

in the context in which real-time traffic analysis is needed, MLP is the best as it is the only one used in this work that allows an online update.

5.2. Feature importance analysis

Feature selection algorithms aim to select, according to some criterion, a subset from the original set of problem attributes by removing irrelevant or redundant attributes in order to maintain the same, or nearly the same, results (Villela et al., 2011). As the models analyzed in this work already have almost optimal performance, selecting attributes mainly reduces the computational cost of the model. This process also aids us on the analyzes the most critical attributes of the problem.

In this work, we use the Recursive Feature Elimination (RFE) method that works by recursively removing a fixed number of attributes and retraining the model. We used stratified cross-validation with 10-fold to assess the quality of the subsets generated by the RFE. Thus, the subset with the highest accuracy and the lowest number of attributes is selected at the end of the execution. Due to the results, we show in Section 5.1, we decided to use the *Random Forest* model as the internal classifier of the RFE. Another reason for using RF is that it also allows knowing the importance of attributes, called Gini importance, after training the model. Thus, after selecting the features, we also analyze the essential attributes considering the new attributes inserted in the dataset. As the purpose of this work is not to classify the traffic in real-time, there is no problem in keeping attributes that can only be obtained at the end of the flow, such as the *Flow Duration*.

Due to the random nature of RFE, each run may generate a subset of different attributes. Consequently, The resulting subset should be con-

sidered an approximation of the optimal subset of attributes. In Table 7 we summarize the results. The attributes were selected based on the 10-fold, and a validation set was also separated with 33% of the samples with stratified random sampling. It is possible to see that there was a significant reduction in the number of attributes of the *dataset* without any loss in classification accuracy in both tasks. In the first one, for

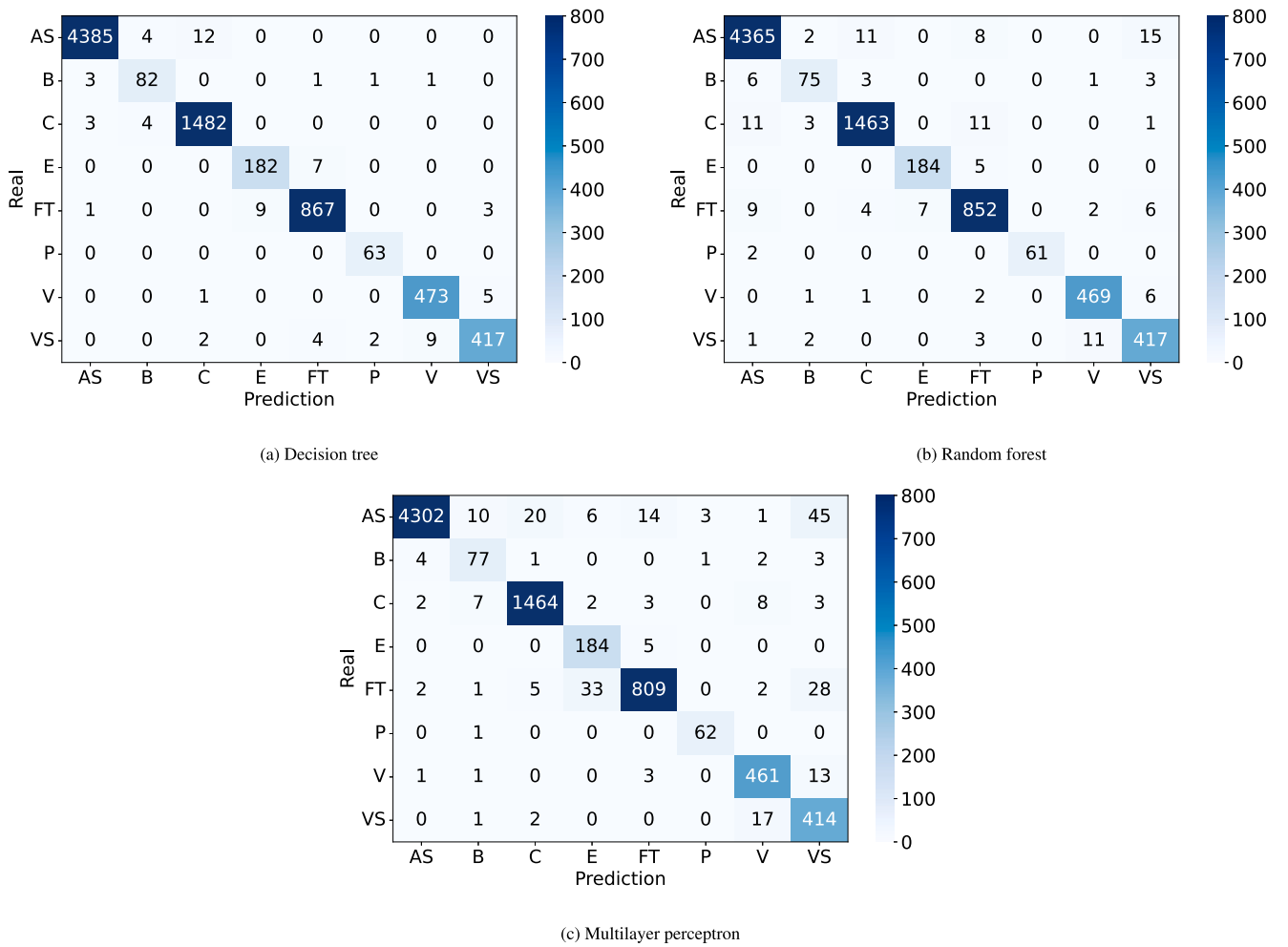


Fig. 10. Models confusion matrix for Darknet traffic detection.

instance, we obtained a reduction of 83% of the total attributes, and in the traffic characterization, the number of attributes is reduced by 72%.

Tables 8 and 9 show the twenty-two most important attributes in the sets selected by RFE in both classification tasks. It is evident that in both sets, the attributes inserted by data pre-processing are in the first place concerning their importance for the models' classification. As we can see, some of the new features have higher ranks than the original features, for example, the features `col_91` and `col_76` have, respectively, relevances of 0.7628 and 0.4287 in their respective tasks. This indicates that the new attributes may encode pieces of information that are relevant to their tasks and that it is more advantageous to process the attributes than to remove them when they do not appear to be relevant, as done in Gurdip Kaur and Habibi Lashkari (2020).

6. Conclusion

In this work, we approach the problems of detection and characterization of traffic coming from *Darknet* through the use of learning models based on decision trees, being DT, RF, and MLP capable of classifying new records of traffic with an accuracy greater than 97% for each of the classification tasks.

New features were also extracted from the original *dataset* by searching for information on the source and destination IPs of the traffic and encoding them with *hashing encoding*. Another feature generated was the time when the traffic occurred by the *timestamp* included in the *dataset*. Our initial analysis showed the potential to contribute to the efficiency of the trained models due to the trend of occurrence of internet traffic common and *Darknet*, at different times. Finally, we also

performed a feature selection process with the RFE algorithm. We verified that the newly generated features are relevant for the prediction of the models. This makes it evident that, in some cases, it is preferable to process features that, at first sight, have little relevance instead of just removing them. Furthermore, it was possible to obtain a significant reduction in the number of attributes of the original *dataset* without losing prediction performance.

It is evident that simple machine learning algorithms, such as decision tree-based ones, are good candidates for obtaining competitive results for real-world problems. In this work, we observed that DT, RF, and MLP obtained a result up to 13% higher than the model presented in Gurdip Kaur and Habibi Lashkari (2020), in addition to its efficiency being improved with a careful pre-processing of existing attributes.

Through McNemar's statistical test, it was possible to observe that all models obtained equivalent performance for the traffic detection task. However, for the traffic characterization, the best model was the DT, and the MLP had the worst performance concerning the unbalanced labels in the *dataset*. Although MLP has the worst performance, it is the most suitable for applications that need to analyze traffic in real-time as it can be updated as new examples arrive. In future work, it is necessary to analyze techniques to improve the performance of MLP concerning unbalanced labels through methods, for example, that generate artificial data for the minority classes.

As future research directions are possible to conduct experiments to classify flows inside cryptographic tunnels. A candidate to help in this way is the framework presented by Bovenzi et al. (2020). Another way is to involve Deep Packet Inspection (DPI) as enrichment for data analysis.

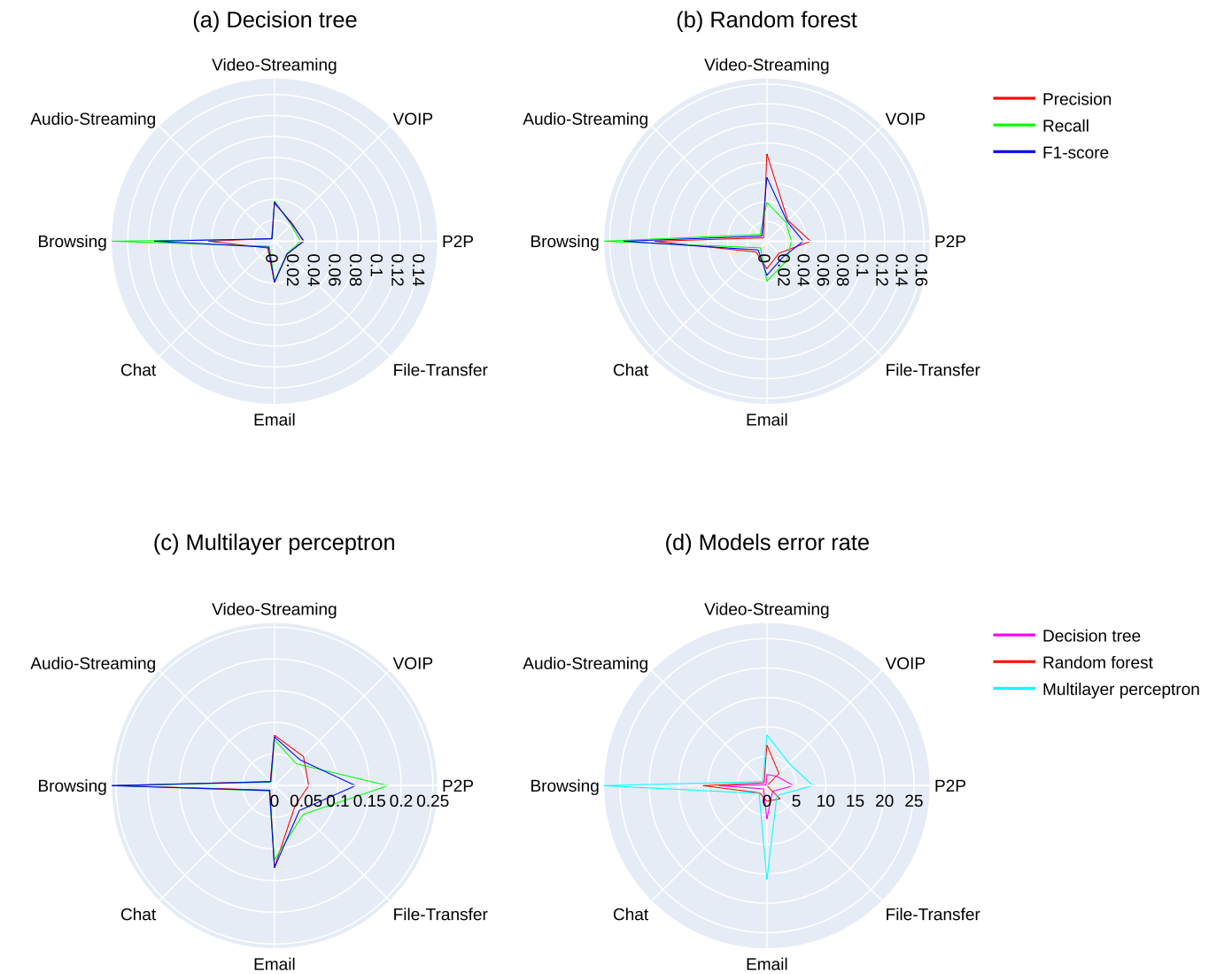


Fig. 11. Evaluation metrics for the models in polar coordinates.

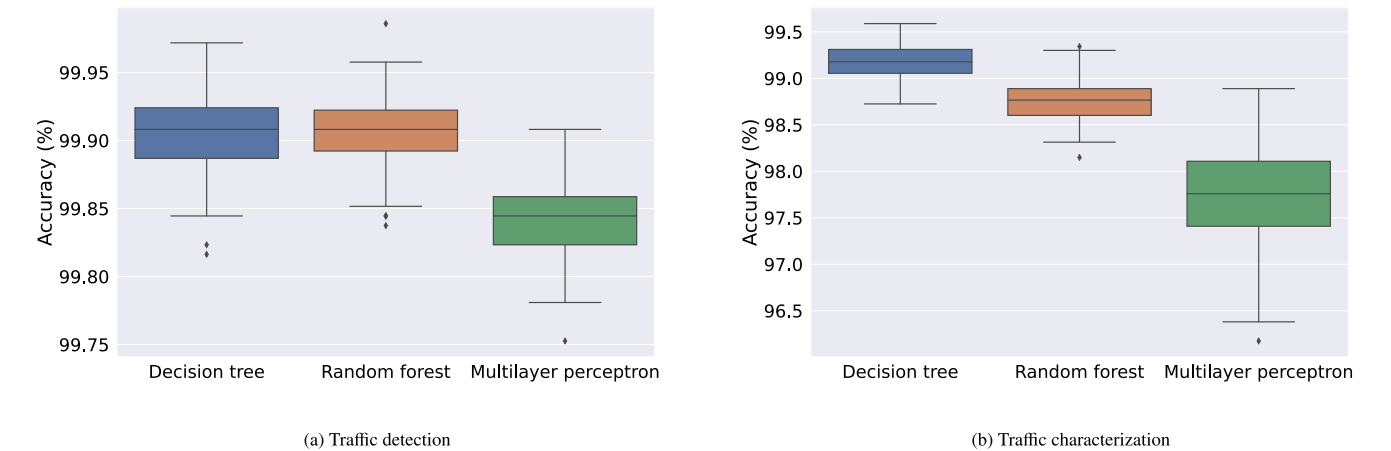


Fig. 12. Boxplot of accuracies estimated by 10-10-fold.

7. List of acronyms		FTP	File Transfer Protocol.....	2
		HTTP	Hypertext Transfer Protocol.....	2
DL	Deep Learning.....	IMAP	Internet Message Access Protocol.....	2
DPI	Deep Packet Inspection.....	IP	Internet Protocol.....	2
DT	Decision Tree.....	KNN	K-Nearest Neighbors.....	2

Table 9
Traffic detection.

Attribute	Relevance
col_76	0.4287
hour	0.1455
Bwd Packet Length Min	0.1262
Idle Max	0.0517
Fwd Header Length	0.0338
Idle Min	0.0335
col_58	0.0312
Packet Length Max	0.0245
Flow Duration	0.0158
col_75	0.0142
col_11	0.0128
col_21	0.0112
col_45	0.0107
Src Port	0.0083
Dst Port	0.0078
Flow IAT Max	0.0053
Fwd Seg Size Min	0.0042
Flow IAT Min	0.0039
col_91	0.0037
FWD Init Win Bytes	0.0029
Subflow Fwd Bytes	0.0029
Fwd IAT Max	0.0026

MLP	Multilayer Perceptron.....	2
P2P	Peer-to-Peer	1
POP3	Post Office Protocol.....	2
RF	Random Forest	2
RFE	Recursive Feature Elimination	8
SMTP	Simple Mail Transfer Protocol	2
SSH	Secure Shell	2
TELNET	TELNET Protocol	2
Tor	The Onion Router	2
VPN	Virtual Private Network.....	2

CRedit authorship contribution statement

Mateus Coutinho Marim: Proposal, experiments and text.

Paulo Vitor Barbosa Ramos: Proposal, experiments, and text.

Alex B. Vieira: Review the methodology, the results of the experiments, and the text.

Edelberto Franco Silva: Review the methodology, the results of the experiments, and the text.

Roberto M. de Oliveira: Review the methodology and evaluation conduction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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